

Area of Composite Figures

Lesson 5-5

Name: _____

Date: _____

Class: _____

Key Vocabulary

Level 1 support

Picture first, then the word, then a plain-language meaning. Say each word out loud.

An L-shaped room = a 12×8 rectangle joined to a 6×5 rectangle; total area = $96 + 30 = 126$ sq ft

Composite Figure

A shape made by putting two or more simple shapes together.

Draw a dashed line across the L-shape to split it into two rectangles — now you can find each area separately

Decompose

To break a shape into smaller, simpler shapes.

T-shaped hallway: top rectangle = 30 sq ft, bottom rectangle = 28 sq ft $\rightarrow$ total = $30 + 28 = 58$ sq ft

Add

Add up the areas of the smaller shapes to get the total.

A 14×10 pool with a 6×4 cutout: $140 - 24 = 116$ sq ft of water surface

Subtract

Take away the area of a missing piece from a bigger shape.

Rectangle: $A = l \times w$; Triangle: $A = \frac{1}{2} \times b \times h$; use the right formula for each piece of a composite figure

Formula

A math rule written with symbols.

Key Ideas & Notes

- Your architecture firm is calculating the floor area of an L-shaped room for a client's new home.
- The contractor needs the total area to order the correct amount of flooring.
- The room can be split into two rectangles: one is 12 feet by 8 feet and the other is 6 feet by 5 feet.
- Break each composite figure into simpler shapes. Find the area of each part, then calculate the total area.

Think About It

- What does the shape of the room look like?
- Can you see simpler shapes inside the L-shape?
- What measurements are given for each part?

My Notes

Turn & Talk — Discussion Points

Level 1 support

Level 2

Talk with a partner about each prompt. If you need help getting started, use the Level 1 sentence stems. Ready for more? Try the Level 2 question.

LAUNCH

The L-shaped room splits into a 12 ft by 8 ft rectangle and a 6 ft by 5 ft rectangle. Why is it easier to find the area after you decompose the L into two rectangles?

Level 1 support

Start here: We decompose the L-shape because we already know how to find a rectangle's area.

We decompose the L-shape into ____ because ____.

Descomponemos la forma de L en ____ porque ____.

After I split it, I can find each area using ____.

Después de dividirla, puedo hallar cada área usando ____.

Word bank: composite figure decompose add subtract formula

Listen for: Students explain that the L-shape is hard to measure directly, so breaking it into two rectangles lets them use $A = l \times w$ on each piece and add the results.

Level 2

The wonder prompt asks if there is more than one way to split the L. Show a different decomposition and justify why both give the same total area.

- Another way to split it is ____, which gives ____ + ____.
- Both ways give the same total because ____.

EXPLORE

For the U-shaped pool you subtracted a 6 ft by 4 ft cutout from a 14 ft by 10 ft rectangle. How do you decide when to ADD areas and when to SUBTRACT?

Level 1 support

Start here: We add when shapes are joined and subtract when a piece is removed.

I broke the figure into ____.

Dividí la figura en ____.

I subtracted the areas because a piece was ____, so 140 minus 24 equals ____.

Resté las áreas porque una parte fue ____, entonces 140 menos 24 es ____.

I add areas when ____, but I subtract when ____.

Sumo áreas cuando ____, pero resto cuando ____.

Word bank: composite figure decompose add subtract cutout

Listen for: A strong answer states you subtract when a piece is cut out ($\text{pool} = 140 - 24 = 116 \text{ sq. ft}$) and add when shapes are joined, with a clear reason for each.

Level 2

Could you find the U-shaped pool's area by ADDING rectangles instead of subtracting a cutout? Justify whether both strategies reach the same answer.

- I could add by splitting the pool into ____, which gives ____.
- Both strategies match because ____.

CONNECT

The T-shaped conference room has a top of 14 ft by 4 ft and a stem of 6 ft by 10 ft, and carpet costs \$5 per sq ft. Talk through finding the total cost.

Level 1 support

Start here: Find each rectangle's area, add them, then multiply by the price.

The top area is ____ square feet and the stem area is ____ square feet.

El área de arriba es ____ pies cuadrados y el área del tronco es ____ pies cuadrados.

I added the areas to get ____ square feet.

Sumé las áreas para obtener ____ pies cuadrados.

The cost is ____ times \$5, which equals \$ ____.

El costo es ____ por \$5, que es \$ ____.

Word bank: (composite figure) (decompose) (add) (subtract) (formula)

Listen for: Students compute $top = 56 \text{ sq ft}$, $stem = 60 \text{ sq ft}$, $total = 116 \text{ sq ft}$, and $cost = 116 \times \$5 = \580 , explaining the add-then-multiply order.

Level 2

A house-shaped wall is a rectangle topped with a triangle. Explain how composite-figure thinking handles a figure made of two DIFFERENT shape types, not just rectangles.

- I would find the rectangle area with ____ and the triangle area with ____.
- Then I ____ the areas because ____.

Guided Examples

Example 1

An L-shaped room is made of two rectangles: $10\text{ ft} \times 6\text{ ft}$ and $4\text{ ft} \times 3\text{ ft}$. What is the total area?

Solution: Area 1 = $10 \times 6 = 60$ sq ft. Area 2 = $4 \times 3 = 12$ sq ft. Total = $60 + 12 = 72$ sq ft.

Answer: A. 72 sq ft

Example 2

A rectangular patio is $15\text{ ft} \times 10\text{ ft}$ with a $5\text{ ft} \times 4\text{ ft}$ rectangular flower bed cut out. What is the remaining area?

Solution: Patio = $15 \times 10 = 150$ sq ft. Cutout = $5 \times 4 = 20$ sq ft. Remaining = $150 - 20 = 130$ sq ft.

Answer: A. 130 sq ft

Example 3

Find the area of an L-shape made of a 5×3 rectangle and a 2×4 rectangle.

Solution: $5 \times 3 = 15$. $2 \times 4 = 8$. Total = $15 + 8 = 23$ sq units.

Answer: A. 23 sq units

Write About the Math

The Writing Revolution

I can explain my steps using the words composite figure, decompose, add, and subtract.

1. Kernel Sentence subject + verb

Model: Composite Figure is a shape made by putting two or more simple shapes together.
Figura compuesta es una figura formada al juntar dos o más figuras simples.

Write a kernel sentence about composite figure. Use a subject and a verb.

Escribe una oración base sobre figura compuesta. Usa un sujeto y un verbo.

2. Sentence Expansion because · but · so

Kernel: Composite Figure matters in math
Figura compuesta importa en matemáticas

Expand the kernel three ways. Add a reason, a contrast, and a result.

because
porque

Composite Figure matters in math because ____.
Figura compuesta importa en matemáticas porque ____.

but
pero

Composite Figure matters in math, but ____.
Figura compuesta importa en matemáticas, pero ____.

so
entonces

Composite Figure matters in math, so ____.
Figura compuesta importa en matemáticas, entonces ____.

3. Sentence Types 4 ways to write a math idea

Statement *Afirmación*

Tell one true fact about composite figure.
Di un hecho verdadero sobre composite figure.

Composite figure ____.

Question *Pregunta*

Ask a question about composite figure.
Haz una pregunta sobre composite figure.

How does ____ ?

¿Cómo ____ ?

Exclamation *Exclamación*

Show excitement about composite figure.
Muestra entusiasmo sobre composite figure.

Wow, ____ !

¡Guau, ____ !

Command *Mandato*

Tell a partner what to do with composite figure.
Dile a un compañero qué hacer con composite figure.

First, ____ .

Primero, ____ .

4. Explain Your Reasoning use a sentence starter

We decompose the L-shape into ____ **because** ____.

Descomponemos la forma de L en ____ *porque* ____.

After I split it, I can find each area using ____.

Después de dividirla, puedo hallar cada área usando ____.

I broke the figure into ____.

Dividí la figura en ____.

Try It

Solve on your own. Check the answer key when you are done.

1. Find the area of an L-shape made of a 5×3 rectangle and a 2×4 rectangle.

- A. 23 sq units
- B. 15 sq units
- C. 8 sq units
- D. 20 sq units

Show your work:

2. When should you subtract areas instead of adding them to find the area of a composite figure?

- A. When a piece is cut out or removed from a larger shape
- B. When the shapes overlap
- C. When one shape is larger than the other
- D. Always subtract — never add

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

A T-shaped room can be decomposed in two different ways. Show both ways and prove they give the same total area. Use a T-shape where the top is $14\text{ ft} \times 4\text{ ft}$ and the stem is $6\text{ ft} \times 10\text{ ft}$.

Sentence starter: Way 1: I split it into ____ and ____ . Areas: ____ + ____ = ____ . Way 2: I used a ____ rectangle and subtracted ____ . Areas: ____ - ____ = ____ . Both give ____ sq ft.

Show your work:

Reflect — Exit Ticket

A composite figure is made of a $9\text{ ft} \times 7\text{ ft}$ rectangle and a $3\text{ ft} \times 4\text{ ft}$ rectangle joined together. What is the total area?

- A. 75 sq ft
- B. 63 sq ft
- C. 12 sq ft
- D. 75 ft

Your answer:

Answer Key & Teacher Guide

1. **Try It 1:** A. 23 sq units — $5 \times 3 = 15$. $2 \times 4 = 8$. $Total = 15 + 8 = 23$ sq units.
2. **Try It 2:** A. When a piece is cut out or removed from a larger shape — *Subtract when a piece is removed from a larger shape (like a window cut from a wall). Add when two shapes are joined together (like an L-shape).*
3. **Exit Ticket:** A. 75 sq ft — $Area\ 1 = 9 \times 7 = 63$ sq ft. $Area\ 2 = 3 \times 4 = 12$ sq ft. $Total = 63 + 12 = 75$ square feet.

Writing (TWR) — what to look for

- **Kernel sentence:** A complete sentence needs a subject and a verb. Example: Composite Figure is a shape made by putting two or more simple shapes together.
- **Expansion:** *because* gives a reason, *but* shows a contrast or exception, *so* shows a result. Answers vary; each must keep the kernel idea and add the correct kind of detail.
- **Sentence types:** Statement ends with a period, question with "?", exclamation with "!", and a command starts with an action verb (a "bossy" verb).